ATM_ancsc Application Note

Revision History

Date	Version	Description
September 29, 2021	0.50	Initial version created.

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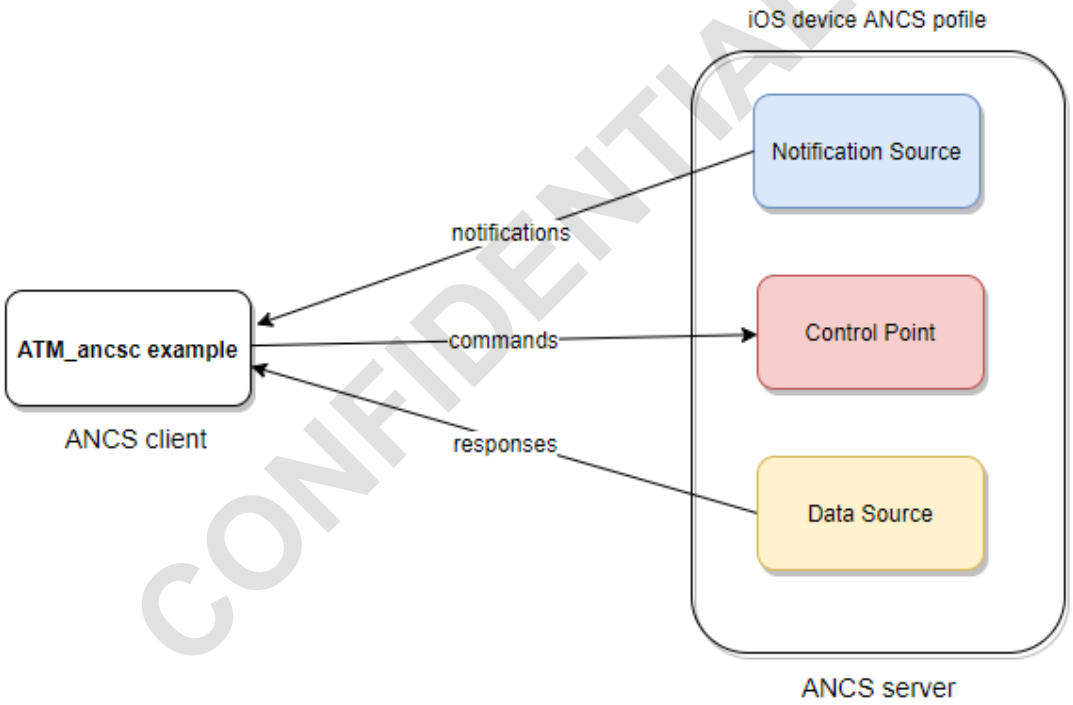
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1 Overview

The Apple Notification Center Service (ANCS) provides Bluetooth accessories a simple and convenient way to access many kinds of notifications that are generated on iOS devices. ATM_ancsc (Atmosic Apple Notification Center Service Client) example is implemented to act as a client role of the ANCS profile and it is able to discover the ANCS server after connecting it with an iPhone. After the service is discovered, ATM_ancsc will enable the Notification Source and Data Source to receive notifications from the server. It could also get more information or perform an action on specific notification by sending a write command to Control Point (ANCS server) as well.



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Figure 1 - Overview of ATM_ancsc example

2 Quick Start

1. To build and run ATM_ancsc
   Go to /examples/ATM_ancsc/ and type the following command

```
make run_all USER_BD_ADDR="a1 a2 a3 a4 a5 a6"
```

2. Find and connect to the device from iPhone Bluetooth Settings
   Connect and pair to "Atmosic 101"

   In the debug log, the Data Source and Notification Source are turned on successfully.

```
@0014ad35 [   app_gap][N]: app_pair_rsp: MITM
@0014addd [   app_gap][N]: app_pair_rsp: Bonding
@0014ae8d [   app_gap][N]: app_pair_rsp: Both using Bonding
@00157e29 [ ATM_ancsc][D]: Pair success
@00157eca ble_get_handler - Unhandled msgid 0xe15 from 0xff
@00157f82 ble_get_handler - Unhandled msgid 0xe15 from 0xff
@0015d6e7 [ ATM_ancsc][D]: Task enable success !!!
@0015e068 [ ATM_ancsc][D]: Data Source Turned On
@0015e6ce [ ATM_ancsc][D]: Notification Source Turned On
```

   Messages and their categories in the log if ATM_ancsc receives notifications from the iPhone.

```
@00317d14 [ ATM_ancsc][D]: Ntf: other (2) added
@00317ddb [ ATM_ancsc][D]: Ntf: other (1) removed
@007f3fb3 [ ATM_ancsc][D]: Ntf: news (167) added
@007f42e6 [ ATM_ancsc][D]: Ntf: news (166) removed
@00844faa [ ATM_ancsc][D]: Ntf: other (0) removed
@00845611 [ ATM_ancsc][D]: Ntf: other (255) removed
@00845fba [ ATM_ancsc][D]: Ntf: other (254) removed
```

3. Make a phone call to the iPhone
   To demonstrate that ATM_ancsc can perform an action to the iPhone, it will reject any incoming call by default. The incoming call id can be seen in the log.

```
@027985ec [ ATM_ancsc][D]: Ntf: incoming call (1) added
@02798c6c [ ATM_ancsc][D]: Call ID: Atmosic
@02798d10 [ ATM_ancsc][D]: reject incoming call
@027992c2 [ ATM_ancsc][D]: Ntf: incoming call (0) removed
@027a491e [ ATM_ancsc][D]: Ntf: missed call (1) added
```

3 State Machine

Figure 2 shows the state machine of this example. The state machine of ATM_ancsc starts from INIT state and then it will try to init BLE, ATM modules first. After the initialization is done, the state transfers from INIT to IDLE state. To enable the BLE advertisement for device discovery, the application will create and start an advertising activity triggering the state machine to turn into ADV_STARTING state. When the application gets the advertisement start confirmation from the atm_adv module, the state will change to ADV_STARTED and wait for the connection indication sent from the remote device. Once the application receives the connection request, it will accept the connection by default and the state will transfer to the CONNECTED state. In the CONNECTED state, the application searches for the ANCS profile and tries to enable the Notification Source and Data Source if the ANCS profile is found. After the two notification sources are enabled, it means that the application is able to receive the notifications from the iOS device. Finally, if the connection is lost or terminated, the application will restart the advertisement again and wait for the next connection indication.

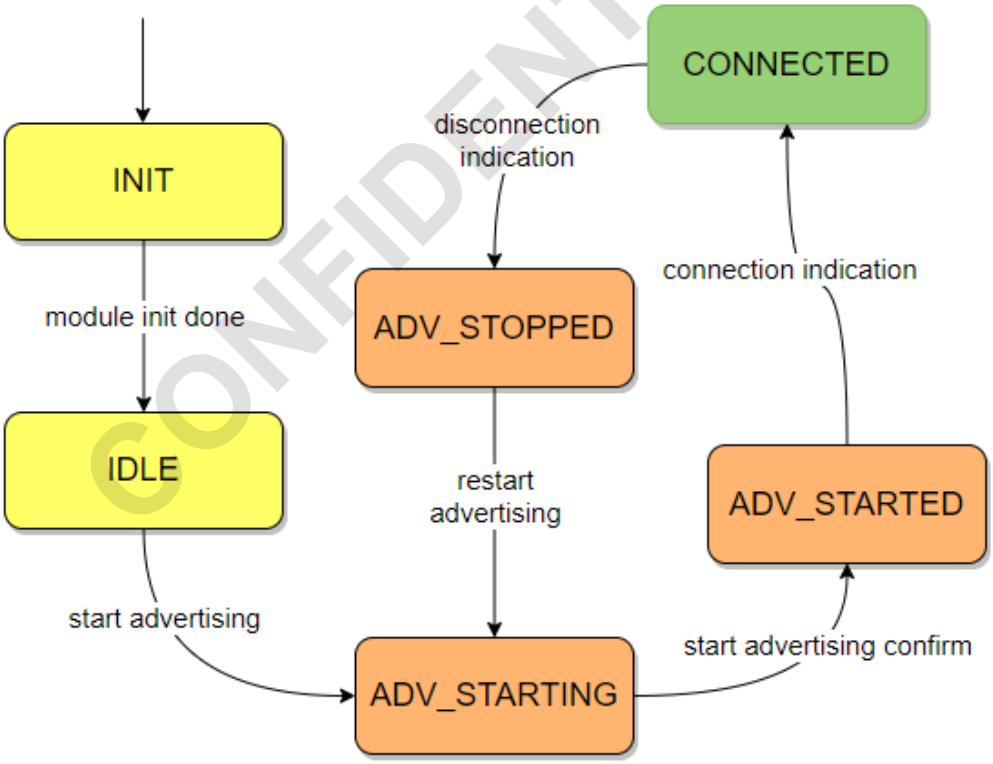


Figure 2 - State Machine of ATM_ancsc Example

4 Hierarchy and Dependency

Figure 3 shows the hierarchy of the ATM_ancsc example. Several BLE and ATM modules are included in this example. The atm_adv and atm_gap module is used for BLE advertisement and

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connection. The ble_ancsc and ble_gattc modules are responsible for searching and enabling the ANCS profile.

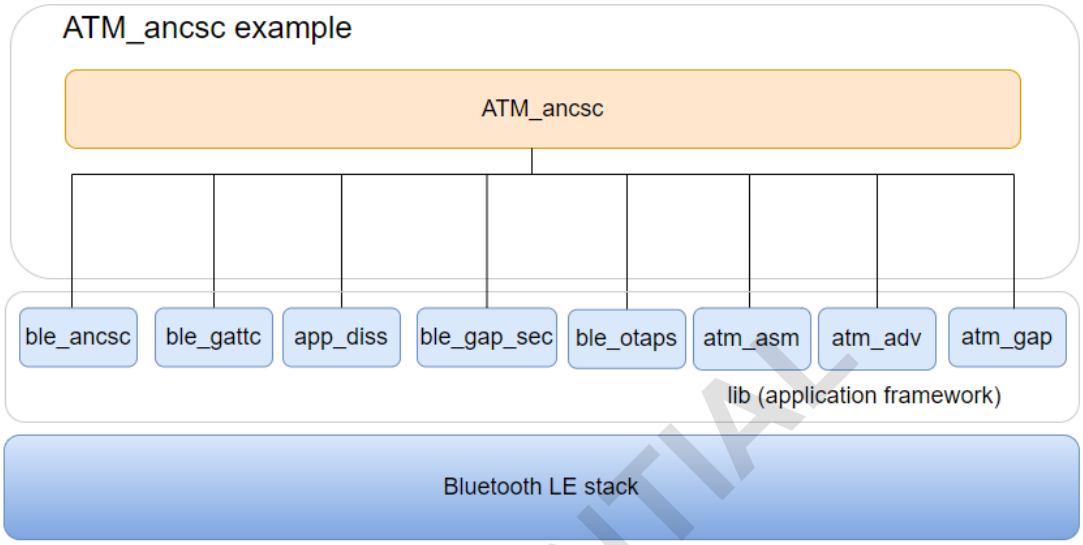


Figure 3 - Hierarchy of ATM_ancsc Example

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5 Parameters

5.1 Advertisement Parameters

The related configuration of BLE advertisement can be found in the atm_ancsc_param_adv.h. The user can easily make their own configuration by modifying this header file.

5.1.1 Min/Max Interval (50 ms)

```
#define ADV0_INTERVAL_MIN 50
#define ADV0_INTERVAL_MAX 50
```

5.1.2 Data (Name: Atmosic101)

```
#define ATM_ANCSC_ADV_DAT_UUID 0x03, 0x03, 0x0D, 0x18
#define ATM_ANCSC_ADV_DAT_APPEARANCE 0x03, 0x19, 0xC0, 0x00
#define ATM_ANCSC_ADV_DAT_NAME 0x0C, 0x09, 'A', 't', 'm', 'o', 's', 'i',
'c', \
     ' ', '1', '0', '1'
#define CFG_ADV0_DATA_ADV_PAYLOAD ATM_ANCSC_ADV_DAT_UUID, \
```

```
    ATM_ANCSC_ADV_DAT_APPEARANCE, ATM_ANCSC_ADV_DAT_NAME \
```

5.1.3 Max Tx power (0 dbm)

```
#define CFG_ADV0_CREATE_MAX_TX_POWER 0x00
```

5.1.4 Duration (always)

```
#define CFG_ADV0_START_DURATION 0
#define CFG_ADV0_START_MAX_ADV_EVENT 0
```

5.2 GAP Parameters

The related GAP parameters of this example can be found in atm_ancsc_param_gap.h header file.

5.2.1 Name (Atmosic 101)

```
#define CFG_GAP_DEV_NAME "Atmosic 101"
```

5.2.2 Suggest Min/Max connection interval (30 ms)

```
// unit: 1.25 ms
#define CFG_GAP_CONN_INT_MIN 24
#define CFG_GAP_CONN_INT_MAX 24
```

5.2.3 Suggest supervision timeout (5 s)

```
// unit: 10 ms
#define CFG_GAP_CONN_TIMEOUT 500
```

5.2.4 Suggest slave latency (no slave latency)

```
#define CFG_GAP_SLAVE_LATENCY 0
```

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6 Bluetooth LE ANCSC Module API Involved

In order to search and communicate with an ANCS profile with the remote side, the ble_ancsc module is needed in this example. The ANCS client APIs involved in this application can be seen in the ble_ancsc.h header file.

6.1 Search for ANCS Profile (ble_ancsc_enble_req)

```
void ble_ancsc_enable_req(uint8_t conidx, ble_ancsc_enable_type_t type,
ble_ancsc_ancs_info_t const *bonded_info);
```

After establishing a connection with a remote device, the application invokes this API to search the ANCS profile. The search result can be checked from enable_rsp callback specified in the ancsc_param. (see atm_ancsc_enable_rsp function) The user can also specify the bonded_info obtained from the previous connection so that the ble_ancsc module won't search the profile again.

6.2 Enable ANCS Notification Sources (ble_ancsc_subscribe_ntf)

```
void ble_ancsc_subscribe_ntf(uint8_t conidx, ble_ancsc_ntf_src_type_t
ntf_src, bool ntf_on);
```

To receive notifications from the iOS device, the ANCS client should write the cccd of ANCS Notification Source and Data Source. By calling the ble_ancsc_subscribe_ntf API, the application is able to turn on/off the notifications.

6.3 Get Attribute Information from a Notification (ble_ancsc_get_ntf_attr)

```
void ble_ancsc_get_ntf_attr(uint8_t conidx, uint8_t const *ntf_uid, uint8_t
const *attr_id_list, uint16_t attr_id_list_len);
```

In order to get more information about a notification from the server, the ANCS client is able to send a get_attributes command to the ANCS Control Point to get more types of attributes from the server. The user should provide the UID of the notification and attributes list to the API. The application uses this API to get the call title of the notification.

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6.3 Perform Notification Action (ble_ancsc_perform_nft_act)

```
void ble_ancsc_perform_nft_act(uint8_t conidx, uint8_t const *ntf_uid,
ble_ancsc_action_id_t action_id);
```

The ANCS client can perform an action on specific notification by writing a perform-action-command to the server. The user is able to achieve this request by calling ble_ancsc_perform_nft_act API. In this example, the application specifies the UID of the notification and negative action ID to reject a phone call.

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